The Post-API Age

Week 3 · Foundations module · Textbook Chapter 3

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This week at a glance

The web used to invite us in. This week we ask *why* that changed — and how to keep doing public-interest research anyway.

Monday | Concepts

- The golden age, then enclosure, exemption & erosion

Wednesday | Notebook lab

- Measure access instead of just describing it: the access matrix and dead-endpoint forensics

Friday | Textbook revisions

- Propose & peer-review pull requests improving Chapter 3

Companion notebook

`ch-03-post-api.ipynb`

Take-home

The notebook's **Recommended Exercises**: a 7-step guided audit. Fill in the empty cells, run everything, export to HTML, upload to Canvas by **Sunday 11:59pm**.

1. Monday · Concepts

The golden age of web data access

For roughly a decade — about **2008 to 2018** — the web behaved as if it wanted to be studied.

- Twitter opened its API in 2006; Facebook launched the Graph API in 2010; Reddit's API was free and essentially unrestricted; Google exposed trends, maps, and books.
- This was not charity: platforms traded **data access** for **ecosystems**, credibility, and lock-in.
- The bargain built a field — computational social science (Lazer et al. 2009) — and a generation of data journalism and civic tech.

When researchers today say the **"post-API age"** (Freelon 2018), this is the era they measure against.

Built on the bargain

The Wikipedia and Reddit studies that trained a generation — including the instructor's — ran on open APIs, public dumps, and volunteer archives like Pushshift (Baumgartner et al. 2020). No permission, partnership, or payment required.

You will use those same open interfaces in Ch. 10 (Wikipedia).

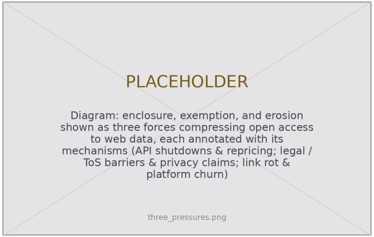
Pressure 1 — Enclosure

Three pressures explain the shift since ~2018: **enclosure, exemption, erosion.**

Enclosure privatizes a shared resource — the name echoes the fencing of English common land. The posts still exist; the *terms of access* are fenced.

- **Twitter/X**: a free academic archive (2021) → discontinued; enterprise access from $42,000/month. Hundreds of tools went dark.
- **Reddit**: free since 2008 → $0.24 per 1,000 calls (2023); Apollo shut down; 7,000+ subreddits went dark; Pushshift was cut off.

The driver: generative AI. Once text became training data, an open API stopped looking like ecosystem-building and started looking like giving away inventory (Reddit–Google: ~$60M/year).



Three forces fencing the open web.

Textbook Ch. 3, "Enclosure." Facebook's CrowdTangle shutdown (2024) was the template.

Pressure 2 — Exemption

Exemption: platforms operate beyond outside scrutiny while running the same studies internally at will.

The asymmetry is the tell. Every A/B test is a behavioral study, and internal teams have total access — but outside researchers meet:

- cease-and-desist letters citing the Computer Fraud and Abuse Act;
- Terms of Service that prohibit the very collection research needs (Fiesler et al. 2020);
- account bans — e.g., Meta disabling NYU's Ad Observatory (2021), justified as *protecting privacy*.

When "privacy" restricts only accountability research — never ad targeting on the same users — it is functioning as exemption, not protection.

Consenting users' data, still shut down.

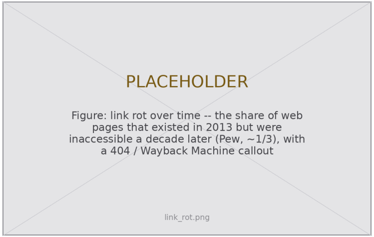
Textbook Ch. 3, "Exemption." Courts push back at the margins (*Sandvig*, *Van Buren*, *hiQ*; see Ch. 2). EU DSA Article 40 may mandate vetted-researcher access.

Pressure 3 — Erosion

Erosion is the quietest pressure: enclosure and exemption are things platforms *do*; erosion is what happens to the web *itself*.

- **Links rot.** About half the links in U.S. Supreme Court opinions are dead; a 2024 Pew study found about a third of 2013 web pages gone a decade later.
- **Platforms churn.** GeoCities vanished; Parler went offline overnight; Twitter became X and broke a decade of embedded links.
- **Sites redesign** and silently break every scraper — fragility you will feel in Ch. 6 & 8.

For research this is a **reproducibility crisis:** a study on the Twitter Academic API cannot be rerun at any price (Davidson et al. 2023).



The record decays unless someone saves it.

Textbook Ch. 3, “Erosion.” Two later chapters are counter-erosion: archives (Ch. 7) and automation (Ch. 14).

This is an old problem — public-interest lineages

Data science is not the first profession to face powerful actors gatekeeping information the public needs. Each precedent offers a hard-won lesson:

- **Journalism** — freedom-of-information laws and the muckraking tradition established that records, public *and* private, are legitimate targets of scrutiny. (You work in this lineage when you file a records request.)
- **Law** — discovery rules compel disclosure; due process demands decisions be explainable and reviewable — inherited almost verbatim by algorithmic accountability.
- **Accounting** — after 1929, mandatory audits turned private finances into public record. Lesson: trustworthy disclosure at scale needs **standards and independent examiners**, not goodwill.
- **Urban planning** — impact statements and public comment periods make community input a precondition for decisions about shared resources.
- **Engineering** — licensure and safety codes established duties to the public that override duties to an employer. Data science is still writing its equivalent.

Textbook Ch. 3, "Public Interest Lineages." Missing Manual Ch. 8: "Artifacts Have Politics."

2. Wednesday · Notebook Lab

Working inside it — openness, oversight, ownership

The counter-values are not a fix for the post-API age; they are a **stance** for working inside it. They frame everything we do today.

Openness

Make your *own* work resist enclosure: transparent pipelines, documented collection, and redundant deposits (Zenodo, Internet Archive, open licenses). Wikipedia is the standing counter-example.

Oversight

Insist consequential systems be inspectable by parties who do *not* own them: audits, documentation standards, DSA Article 40. **The skills this course teaches are oversight skills.**

Ownership

Prefer community-governed infrastructure: federated protocols (ActivityPub, AT Protocol), data cooperatives, ArchiveTeam. The long-horizon answer to enclosure.

Textbook Ch. 3, "Three Counter-Values." Federated platforms return in Ch. 12 (Social).

Claims you can check — the access matrix

Enclosure, exemption, and erosion are claims about the world. Claims about the world can be checked. The chapter's audit asks eight endpoints one question: *if I show up with no credentials, what happens?*

```
def probe(name, url):
    """Ask one endpoint for data without credentials; describe the answer."""
    try:
        response = requests.get(url, headers=HEADERS, timeout=20)
    except requests.RequestException as error: # no answer is a result too
        return {"api": name, "status": None, "verdict": "unreachable"}
    verdict = ("open" if response.status_code == 200
               else "gated" if response.status_code in (401, 403) else "other")
    return {"api": name, "status": response.status_code, "verdict": verdict,
            "content_type": response.headers.get("content-type", "")}
```

Measured 28 Aug 2026: Wikipedia, Hacker News, arXiv, Mastodon, Bluesky **open**; Reddit and X **gated**; Meta Graph refuses without a token. Your run *will* differ — that is the finding, so record the date.

Textbook Ch. 3, "An Access Matrix." Mastodon is the only host that states its rate limit before you hit it.

Read the rows, not the totals

The useful information is in the differences:

- **Wikipedia, Hacker News, arXiv** answer anonymous requests with data, and have for a decade — a non-profit, an incubator, a university consortium. No one here monetizes attention.
- **Reddit**: status 403, content-type `text/html` — not an API refusing you, a *block page* served where JSON lived from 2008 to 2023. Enclosure is a content-type header.
- **X**: `401 application/problem+json` — a machine-readable, honest refusal. Differently built gates tell you whether applying for access would even help.

Attribute before you conclude

While the chapter was drafted, GitHub returned 403 — a striking finding, since GitHub allows 60 anonymous requests/hour. The *body* named the drafting sandbox, not GitHub. Check `server` and `via`, read the body, retest from a second network.

Textbook Ch. 3, "Reading the Failure, Not Just the Status Code." A measurement that describes your instrument is worse than none.

Dead-endpoint forensics — three ways to die

How a retired endpoint refuses tells you whether anything can be recovered:

- **Pushshift** answers `403` in JSON: `{"detail": "Not authenticated"}`. **Gated, not gone** — it still runs, for Reddit moderators. A gated service can be negotiated with.
- **CrowdTangle** does not resolve at all. No HTTP conversation happens. **Enclosure completed**; nothing left to ask.
- **Twitter v1.1** parses your request and rejects it (error 215). Alive, serving paying customers. **Repriced, not removed.**

The archive keeps the receipts

Wayback's last captures: `api.pushshift.io` — 25 May 2023, six weeks before Reddit's pricing took effect. `crowdtangle.com` — 14 Aug 2024, the month Meta retired it. The shutdowns are written in the archival record.

Archived docs are how you reconstruct what a five-year-old paper's dataset contained. More in Ch. 7.

Textbook Ch. 3, "Dead-Endpoint Forensics." Always pass `timeout=` — a dead host may drop packets rather than refuse.

The take-home — a guided audit in seven steps

The notebook's **Recommended Exercises** walk the audit end to end. Fill in each empty cell:

1. Define `HEADERS` with your named User-Agent
2. Probe **one** endpoint by hand; look at the answer
3. Build the eight-endpoint matrix as a DataFrame
4. Add **two** endpoints you might actually study
5. Diagnose one refusal: **`server`**, **`via`**, body
6. Autopsy one retired endpoint: gone, gated, or repriced?
7. Interpret: what differed from the book, and why?

Submitting

Run all cells, then export: *File* → *Save and Export Notebook As* → *HTML*. Upload the **`.html`** file to Canvas by **Sunday 11:59pm**. Canvas accepts only **`.html`** — code and output together, as executed.

Start in class, finish at home. Work in pairs; submit your own run.

Textbook Ch. 3, Recommended Exercises. Date-stamp your matrix — it is a snapshot, not a fact.

If you finish early — the additional exercises

The chapter's **Additional Exercises** are optional, open-ended extensions — worth your time, not required:

1. **Extend the autopsy** — two more retired endpoints: gone, gated, or repriced, and who is left to ask?
2. **Timeline** — five API-access events, three platforms, 2008 to today; tag each with its pressure.
3. **Framework application** — 500 words on a platform change *not* in the chapter.
4. **Contingency plan** — a project on one source, plus three fallbacks if it vanishes mid-study.
5. **Public interest audit** — how would current restrictions hit a real API-enabled project?
6. **5617**: read Davidson et al. (2023) and Perriam et al. (2020); write a 2–3 page research-access brief for one platform.

Fallbacks worth knowing

Official research programs (real but rationed) · data donation · FOIA / records requests (Ch. 11) · crowdsourcing.

Show-and-Tell

Bring one: an access change that surprised you, a dataset you fear losing, or a question for a research design.

Textbook Ch. 3, Additional Exercises & "What This Means for Practice."

Openness in practice — document the collection

The audit teaches you to date-stamp a measurement; openness generalizes that habit. Record **what** you collected and also **how**: endpoints, parameters, dates, and the platform's constraints (Gebru et al. 2021). A future reader (or you, in a year) can then reconstruct exactly what was allowed:

```python
collection = {
    "source": "reddit.com (PRAW, r/ClimateChange)",
    "endpoint": "GET /r/{sub}/top params: t=year, limit=1000",
    "collected": "2026-02-14",
    "access_tier": "researcher program (application-gated)",
    "constraints": "no redistribution of raw comments; 100 req/min",
    "known_gaps": "pre-2023 history unavailable -- Pushshift offline",
}
import json, pathlib
pathlib.Path("datasheet.json").write_text(json.dumps(collection, indent=2))
# The datasheet travels next to the data; the source may change, the record does not.
```

Textbook Ch. 3, "Openness" — documentation as the practice that survives enclosure.

"If the API closes, I'll just scrape."

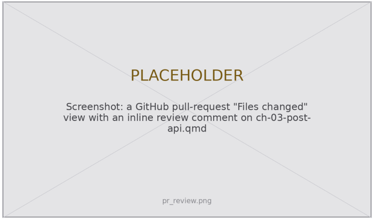
Visibility, permission, and ethics are three different questions.

Scraping is one tool in a portfolio —
not an escape hatch that makes enclosure irrelevant.

3. Friday · Textbook Revisions

Improving Chapter 3 together

Last week you filed an **issue** — a report that something is wrong. Today you take the next step and propose the **fix** itself: your **first pull request**, against `ch-03-post-api.qmd`, plus reviews of two classmates'.



The workflow, in GitHub Desktop or `gh` (no raw `git` needed):

1. **Clone** the book repo, then create a **branch**
2. Edit the chapter; **commit** with a clear message
3. **Push**, then open a PR describing *what* and *why*
4. Review two peers' PRs: kind, specific, actionable

This is a **fast-moving** chapter — prices, lawsuits, and regulations change monthly — so it always needs hands.

Contributions & reviews count toward the Textbook Revisions grade (15%).

A revision menu for Chapter 3

High-value targets — pick one and scope it to a single PR:

- **Add the missing figures.** The chapter still has no figures. Contribute a three-pressures diagram, an API-access timeline (2008–2026), or a link-rot chart — the deck's placeholders show the need.
- **Re-run the audit.** The access matrix and forensics outputs are dated 28 Aug 2026 and *will* drift. Re-run the chapter's code, and where a row changed, update the printed output and the prose that reads it. This is the chapter's built-in revision engine.
- **Update the moving numbers.** Verify and re-cite the Twitter/X pricing, the Reddit–Google deal, and DSA Article 40's implementation status "as of mid-2026." Flag what has gone stale.
- **Resolve the editorial TODOs.** The source carries author-voice and "add citation when published" notes (the openness/oversight/ownership framing). Track down a source or draft the text.
- **Extend a Misconception or Further Reading.** Add a post-2023 source on AI-training licensing, tied to "enclosure logic is spreading."
- **Verify the cross-references.** Ch. 3 leans on many `@sec-` links (ethics, wikipedia, archives, automation, social, government). Confirm each resolves and is described accurately.

Targets drawn from the chapter's callouts, thin exercises, "Common Misconceptions," and "Further Reading."

A good revision, and a good review

A strong PR

- One focused change, clearly titled
- Explains the problem it fixes
- Builds (`quarto render`) without errors
- Matches the book's voice and notation
- Discloses any AI assistance

A strong review

- Runs / reads the change, not just skims
- Names specifics (line, claim, source)
- Separates "must fix" from "nice to have"
- Is generous — assume good faith
- Ends with a clear verdict

Next week: **Data formats** — XML & JSON, the shapes web data actually arrives in.

Key takeaways

1. The open-API era (∼2008–2018) was a **strategic bargain**, not a natural state — and computational social science was built on the trade.
2. **Enclosure** converts collective data into private assets: shutdowns, repricing, and AI-training licensing deals.
3. **Exemption** is the asymmetry of scrutiny — platforms study users freely while blocking those who study platforms.
4. **Erosion** is infrastructural decay, so reproducibility is a problem you must solve *at collection time*.
5. **Access is measurable, not just arguable** — an unauthenticated probe places a platform on the open-to-closed gradient today, and *how* a dead endpoint refuses tells you whether anyone is left to negotiate with.
6. The response is **redundancy**: openness, oversight, ownership — and mastery of many access techniques so no single closure ends your research.

Reading: Textbook Ch. 3. See Freelon (2018); Bruns (2019); Lazer et al. (2020).